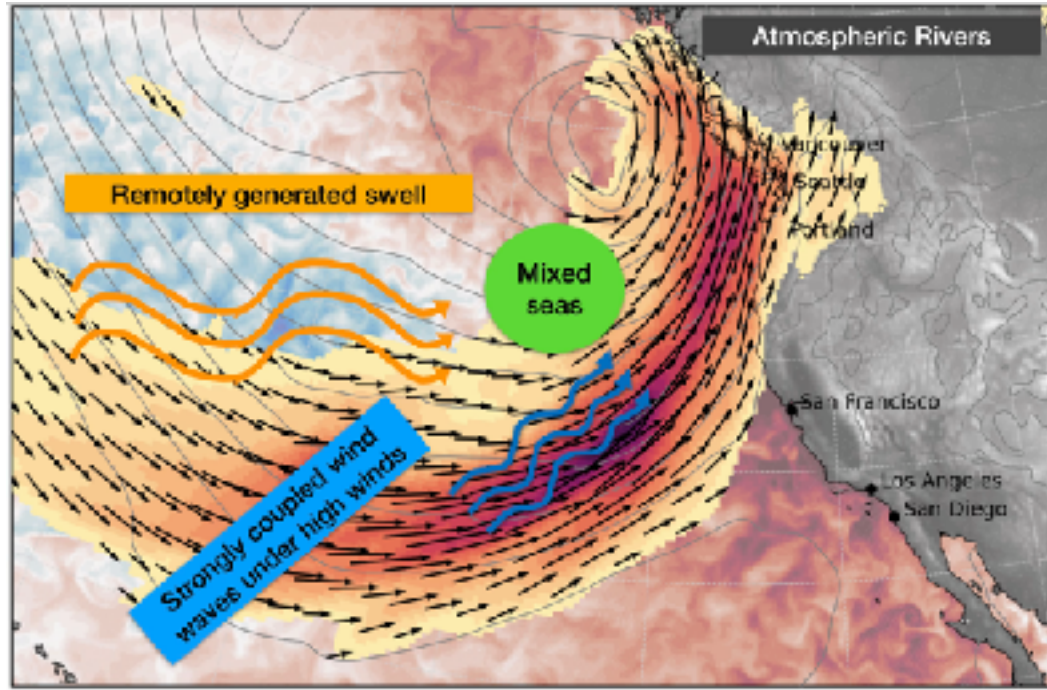
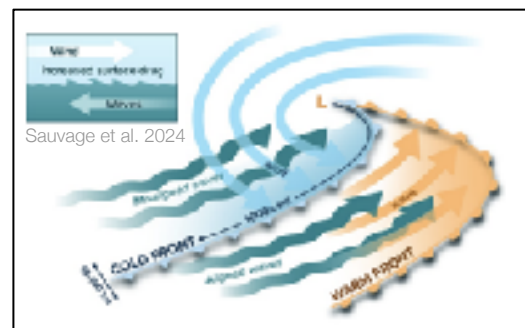
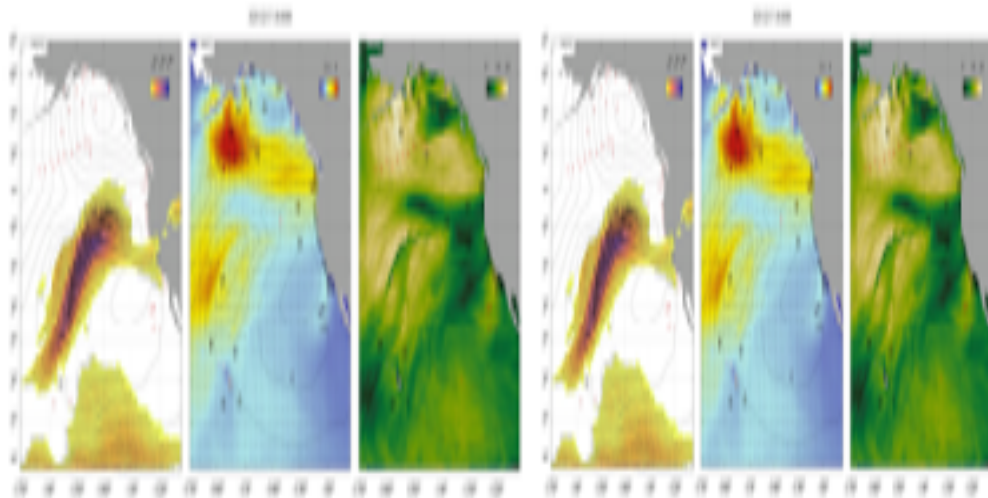


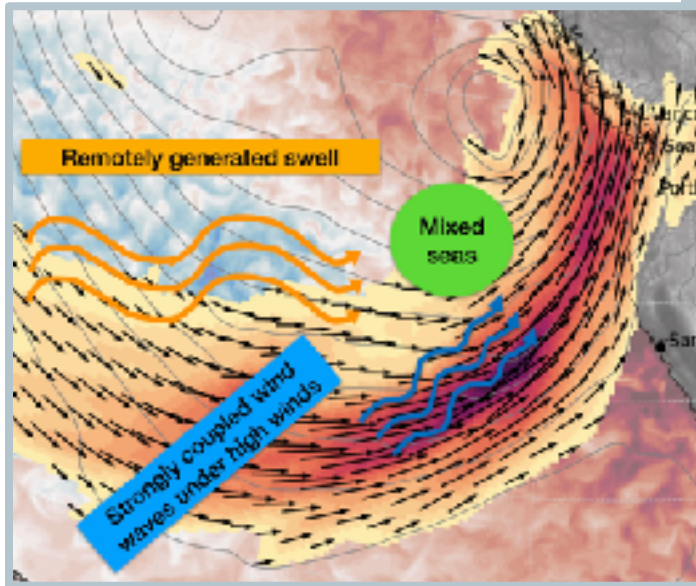
Understanding and Representing Effects of Ocean and Surface Waves in Atmospheric Rivers



Atmospheric rivers at the ocean surface

- maximum IVT followed by cold front (CF) coinciding with
 - rapid reorientation of mean wind direction
 - colder air temperatures
- offset between surface CF and AR: typical time between max IVT and surface CF is 2-6 hours (depends on AR translational speed and orientation)
- CF characterized with substantial misalignment in wake as waves adjust to rapidly varying wind conditions
- misaligned wind-wave conditions shown to change surface stress (e.g. increased drag when waves oppose wind)

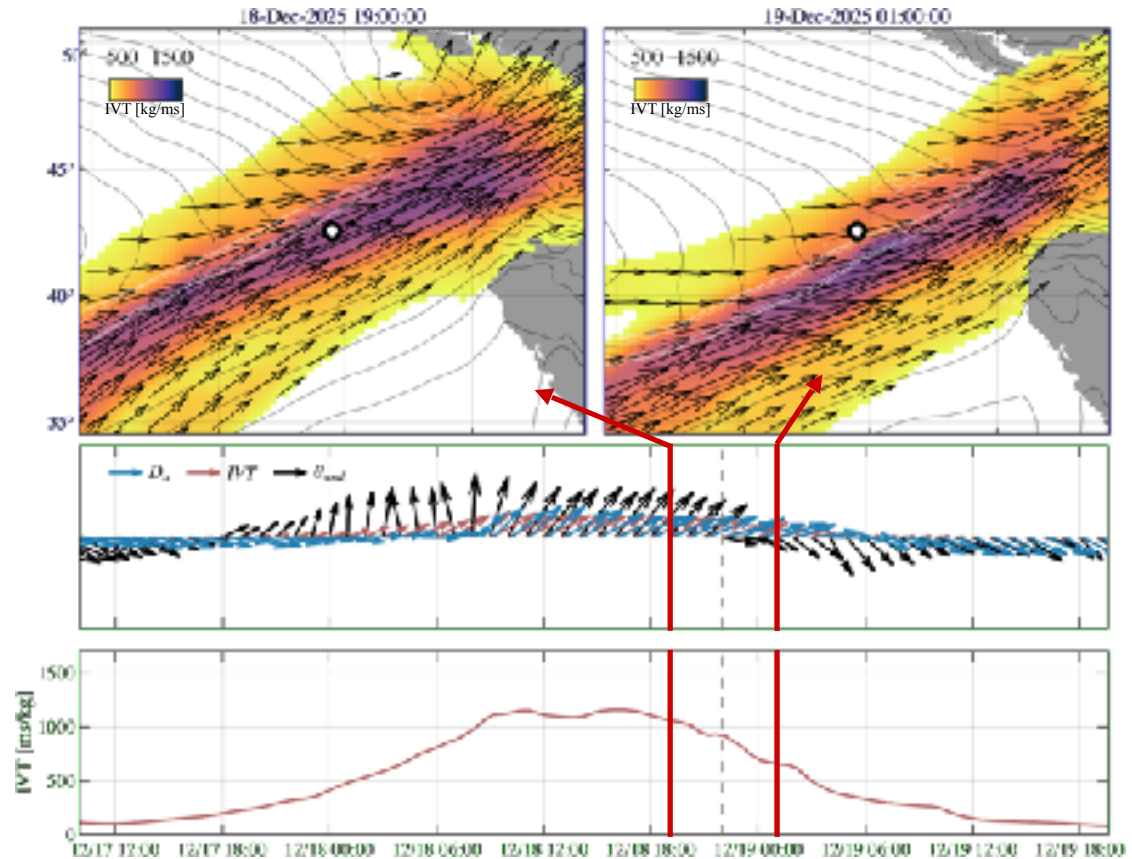
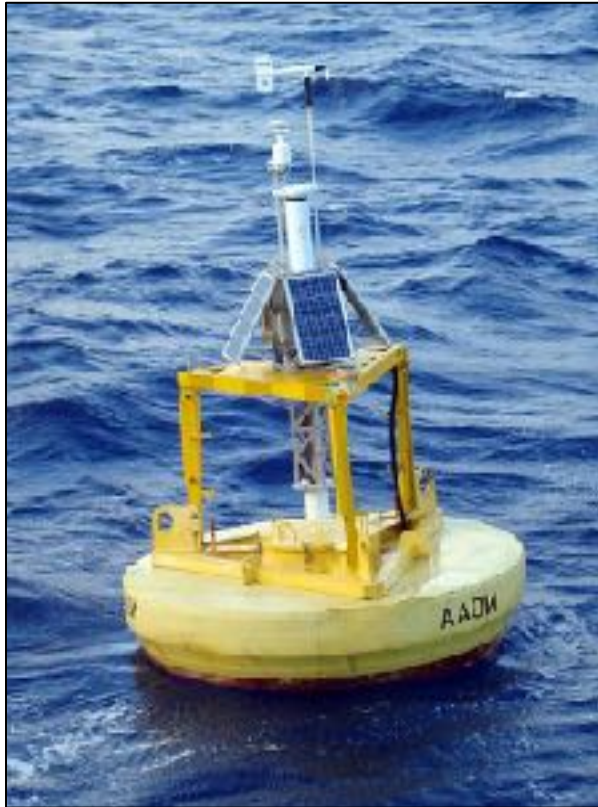




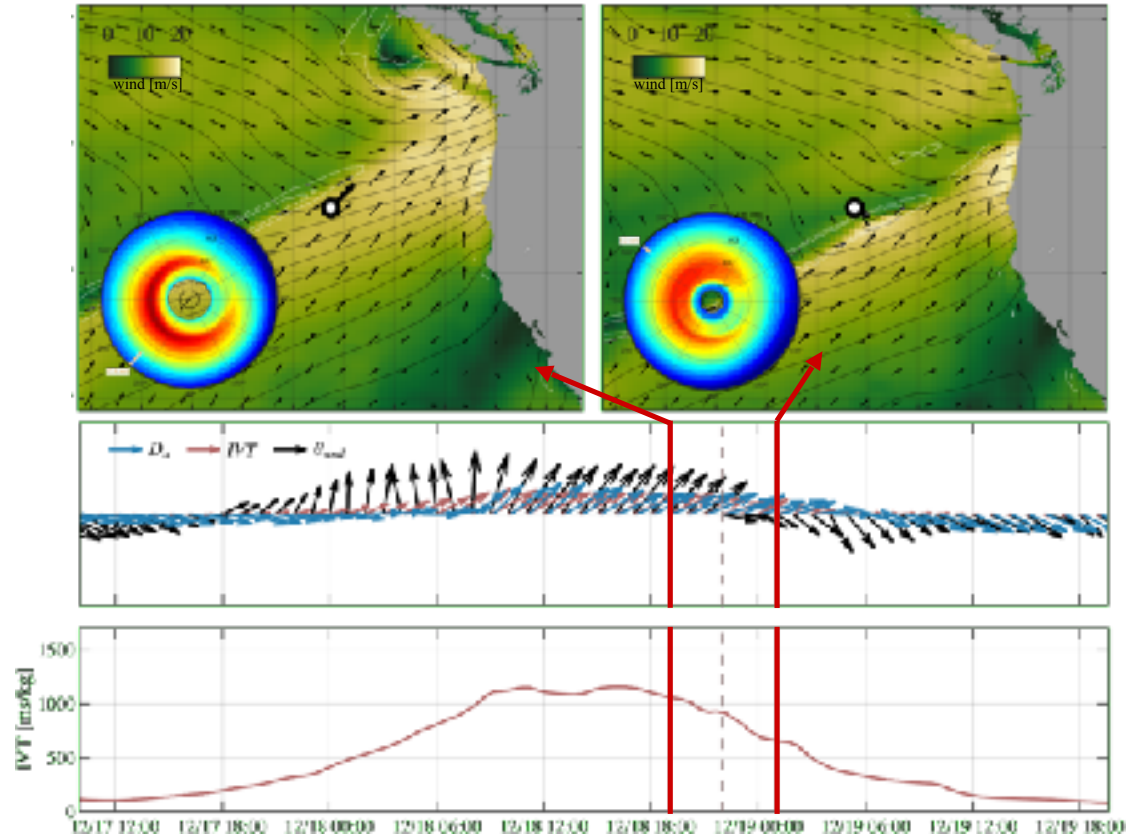
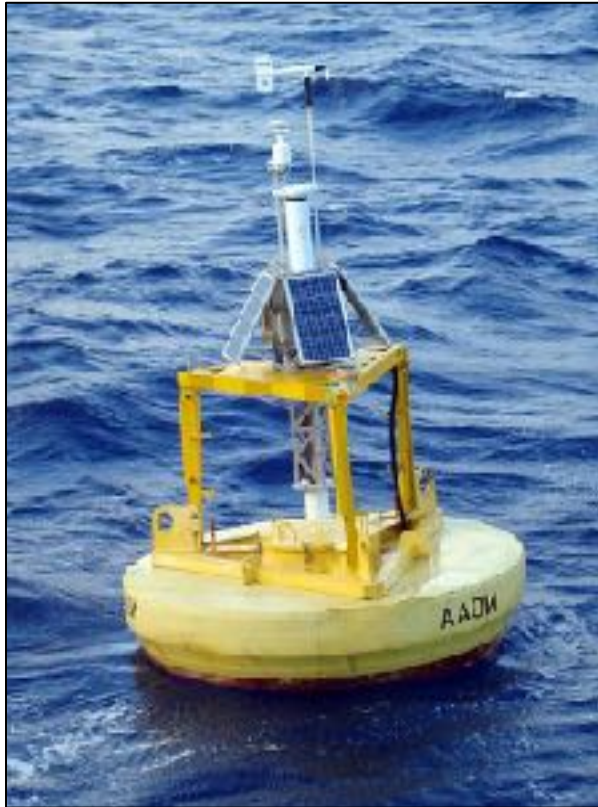
OBJECTIVES

1. characterizing sea states and MBL conditions under ARs in wave and met observations
2. assessing COARE representation of air-sea fluxes within AR, related to sea-state and wind-wave misalignment
3. using SCOAR to study couplings between winds, waves, and currents in targeted AR events

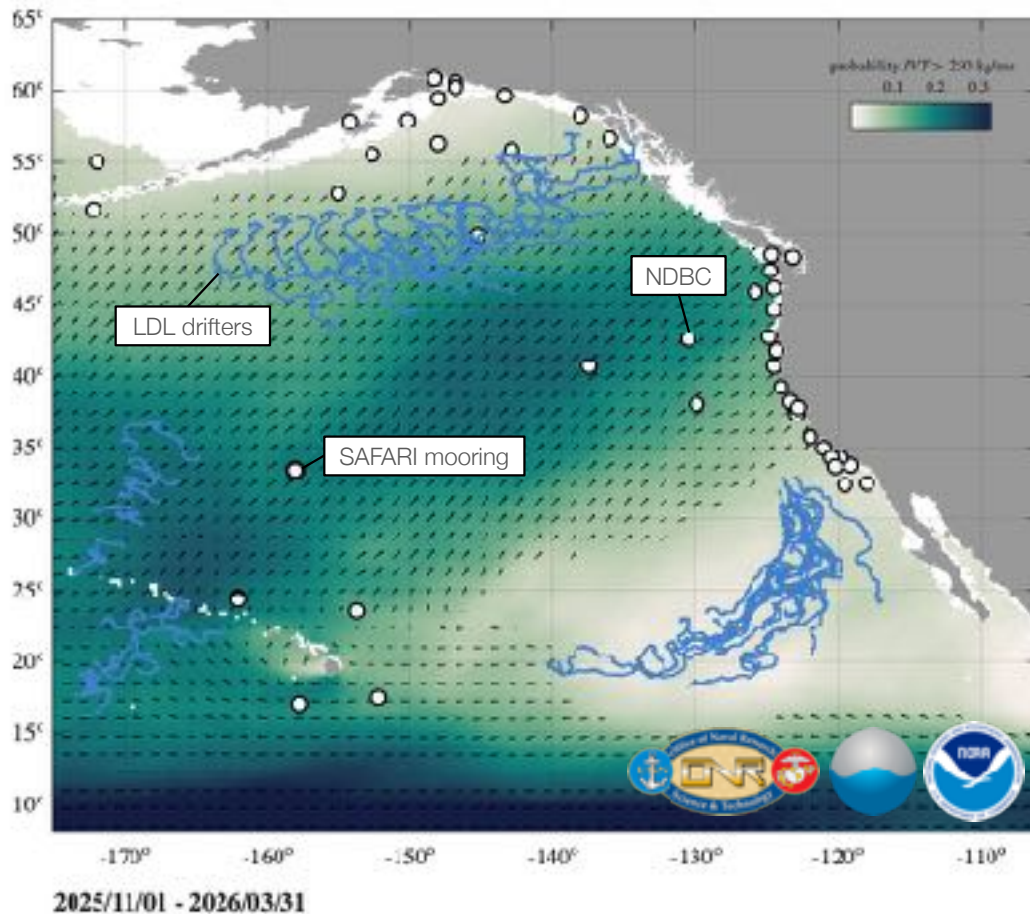
1. observing and characterizing wave conditions in atmospheric rivers



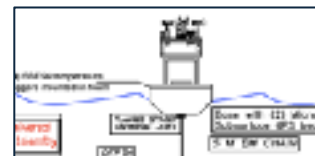
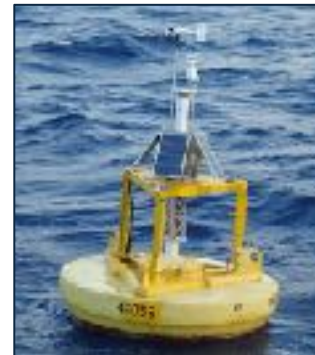
1. observing and characterizing wave conditions in atmospheric rivers



in situ wind and wave wave observations



- SAFARI mooring (wave spectra and mets)
- wave buoys (NDBC) (wave spectra and mets)
- drifters (LDL + SOFAR) (wave spectra)



Thanks to Tom Farrar, Drew Lucas, Martha Schonau, Luca Centurioni, for data access

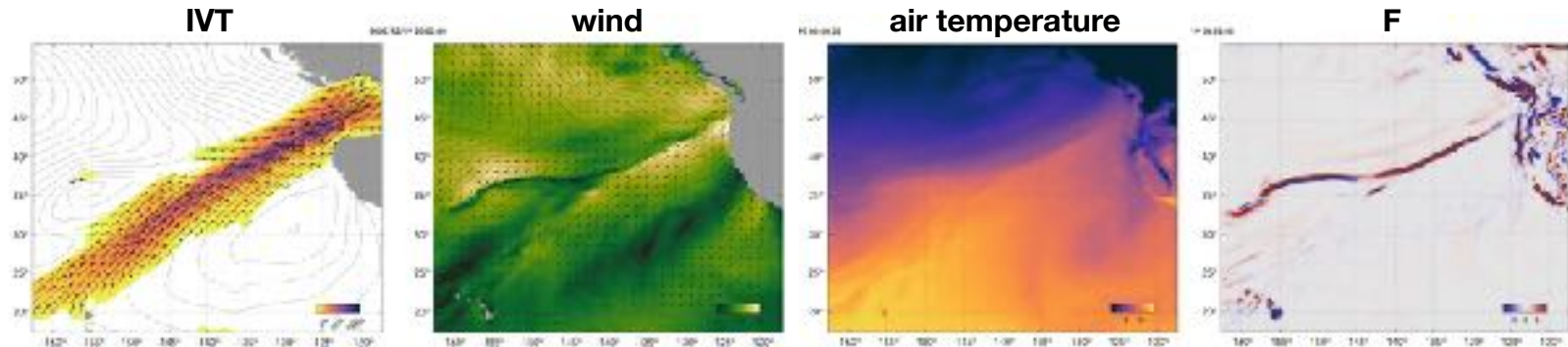
methods

- ERA5 is used to identify fronts

$$F \propto \left| \nabla \times \vec{U} \right| |\nabla T|$$

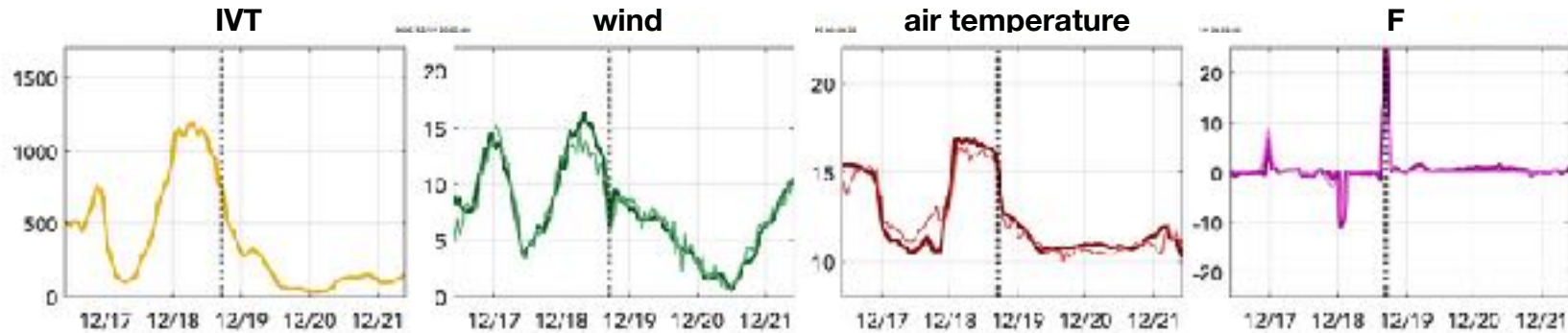
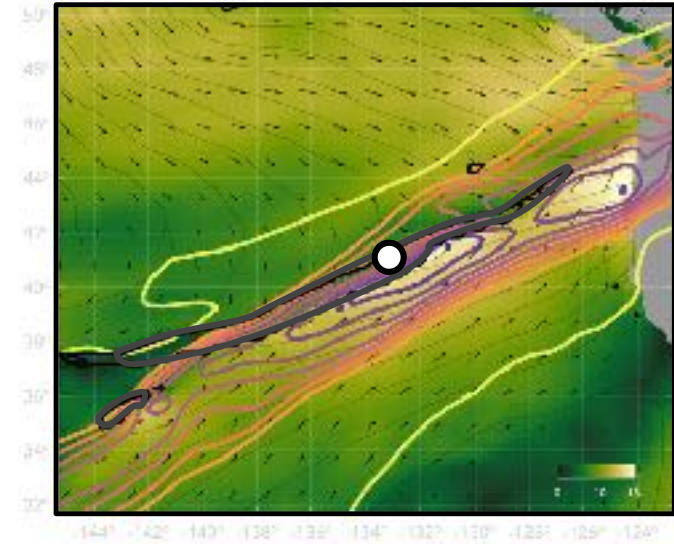
Parfitt, R., A. Czaja, and H. Seo, 2017: A simple diagnostic for the detection of atmospheric fronts, Geophys. Res. Lett., 44, 4351–4358. <https://doi.org/10.1002/2017GL073662>

*** modified at surface*



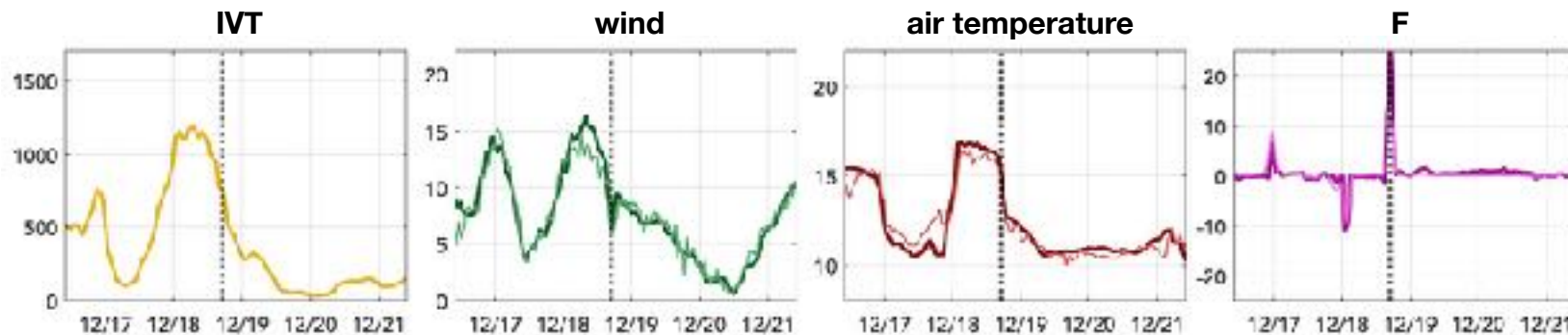
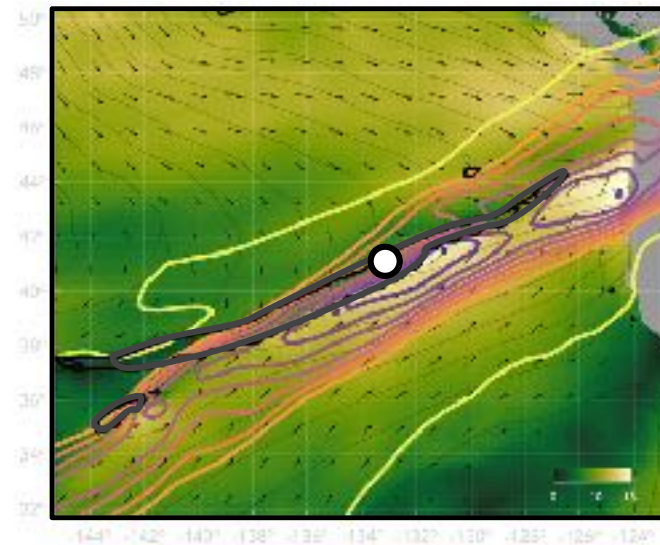
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- ERA5 is used to identify fronts $F \propto \left| \nabla \times \vec{U} \right| |\nabla T|$
- ERA5 is co-located to each buoy observation to provide time series of IVT and F (and winds if not measured)



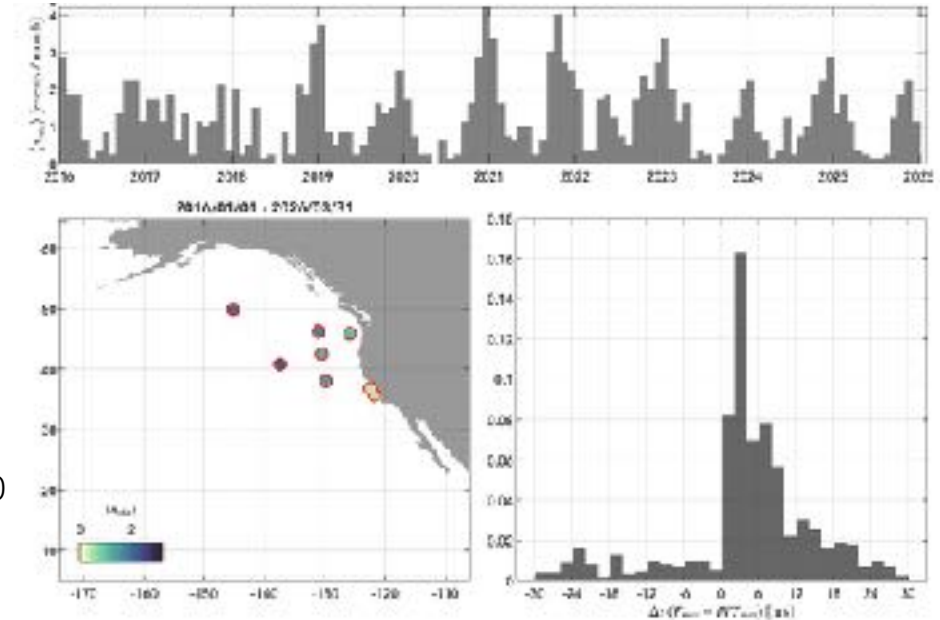
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- ERA5 is co-located to each buoy observation to provide time series of IVT and F (and winds if not measured)
- two criteria are then used to detect AR + CF in observations
 - AR events are identified if IVT > 250 for more than 12 hours
 - CF events are identified if F exceeds a thresholdevents must satisfy both criteria to be selected

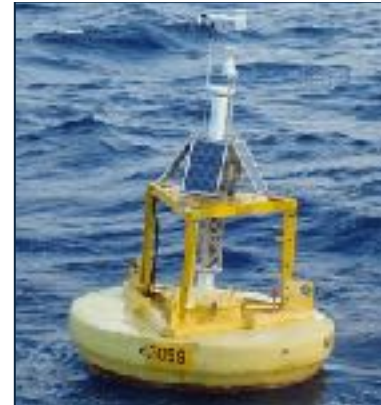
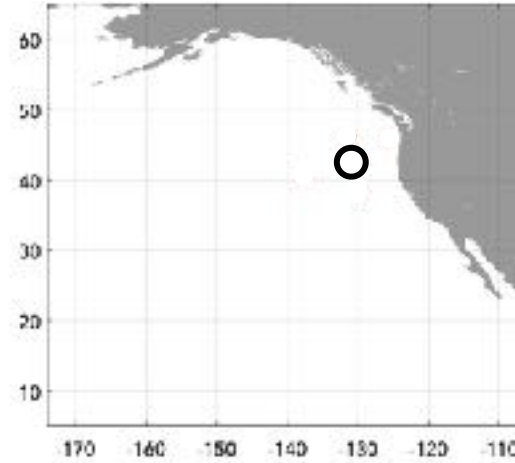
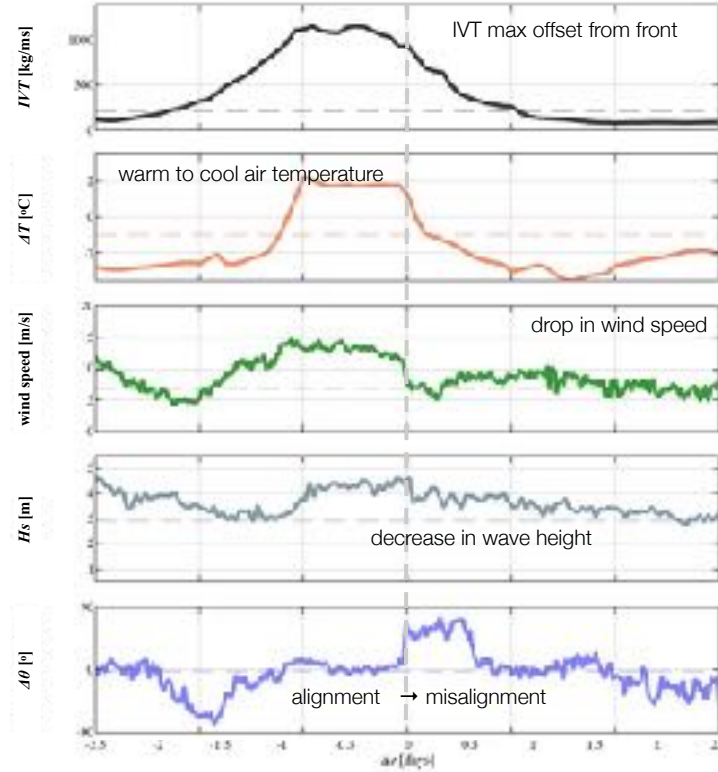


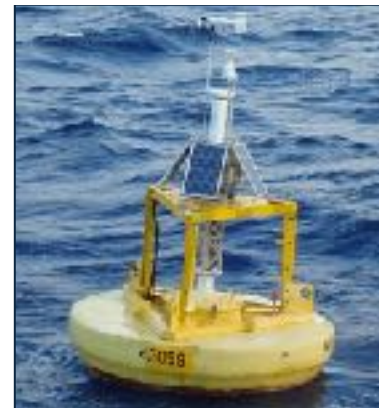
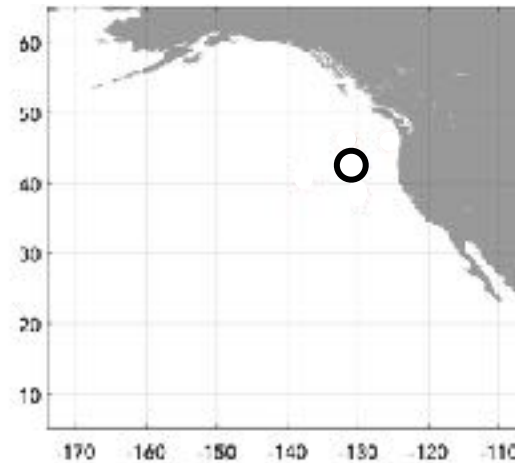
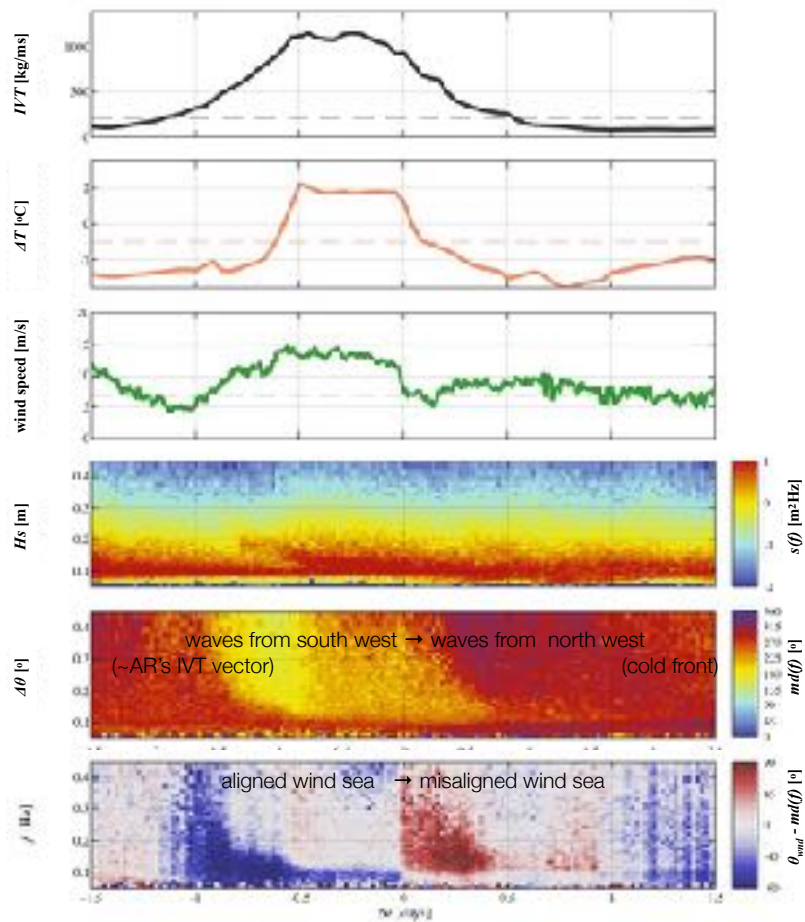
methods

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 - AR events are identified if IVT > 250 for more than 12 hours
 - CF events are identified if F exceeds a thresholdevents must satisfy both criteria to be selected
- observations during each event are re-centered at t=0 when the front passes, based off the maximum turning in the wind

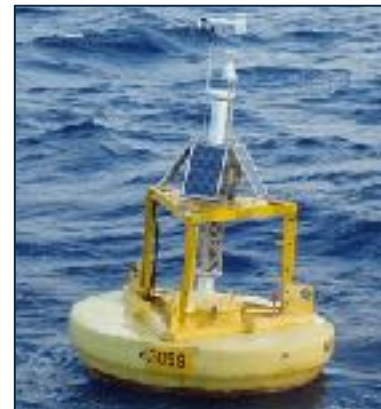
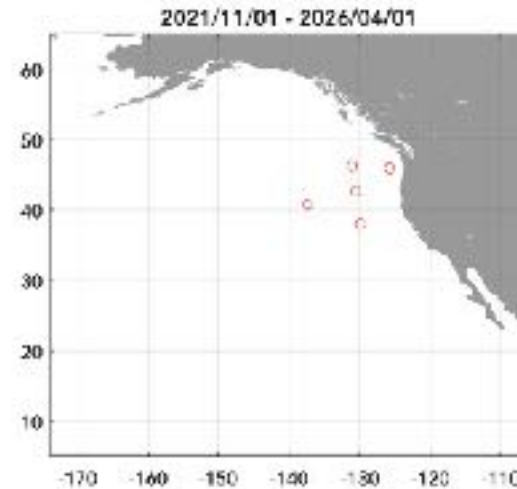
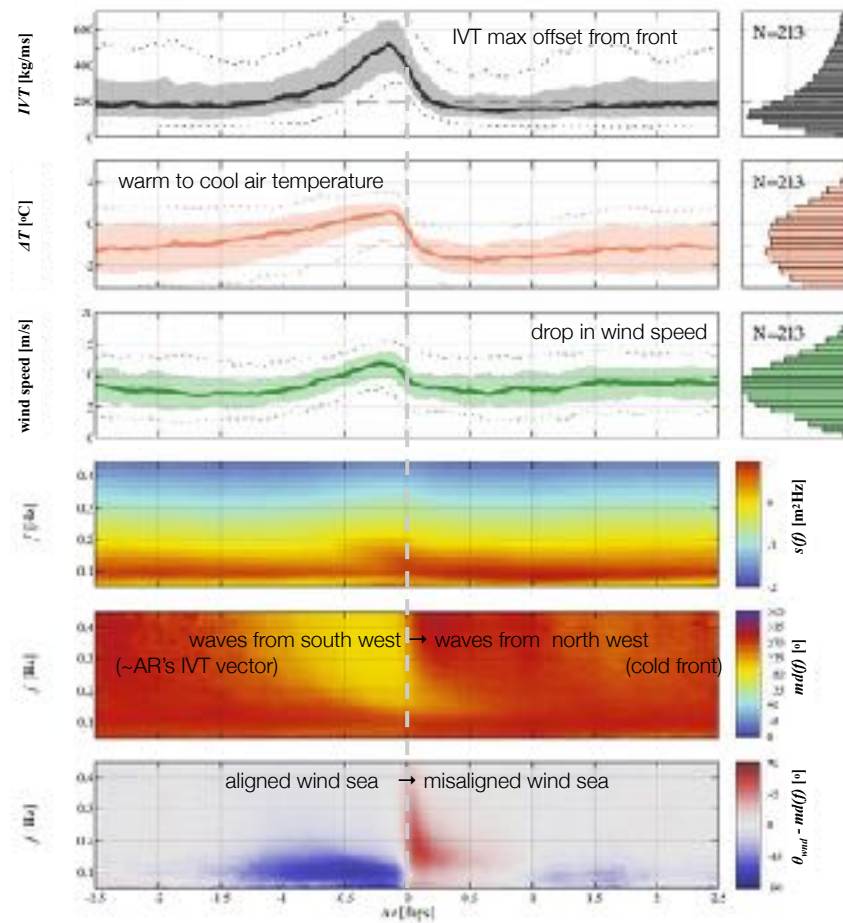


more ARCF events detected during winter; at offshore buoys; median time delay between IVT max and CF is 2-4 hours.

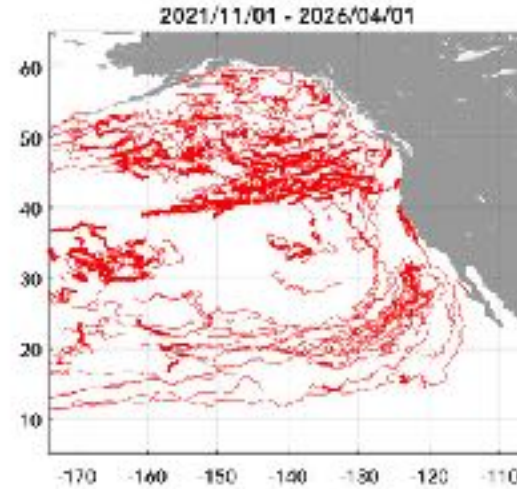
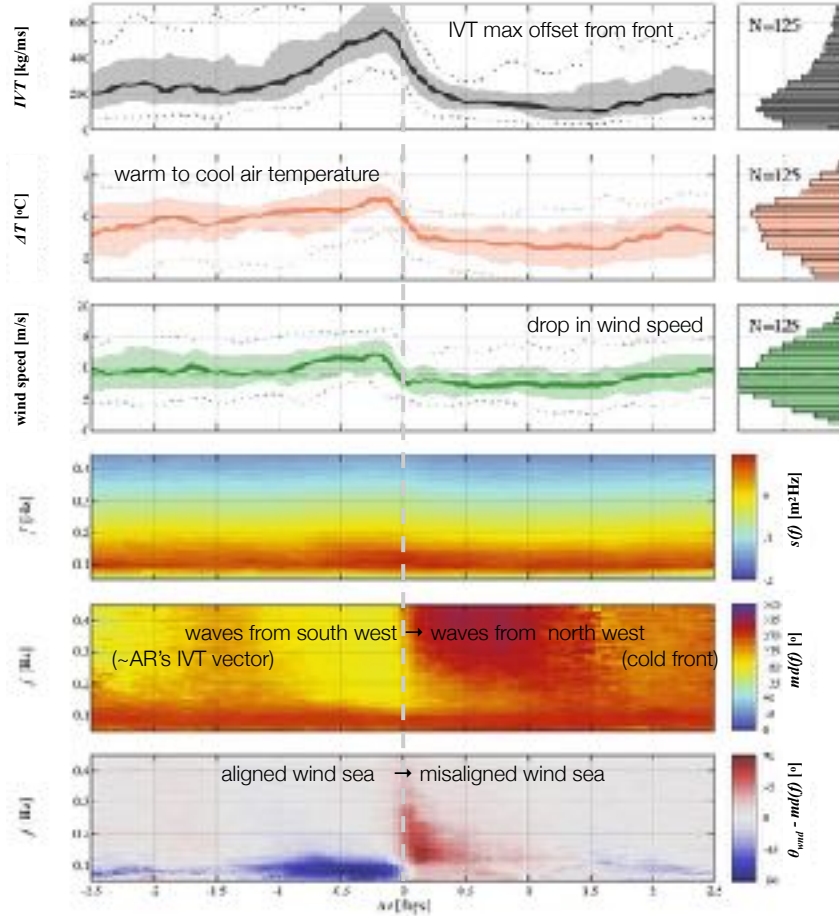




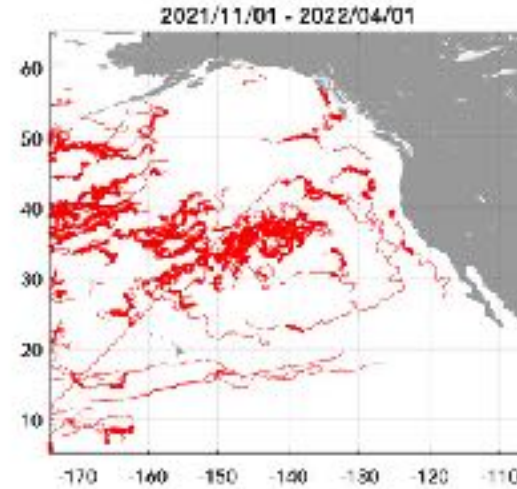
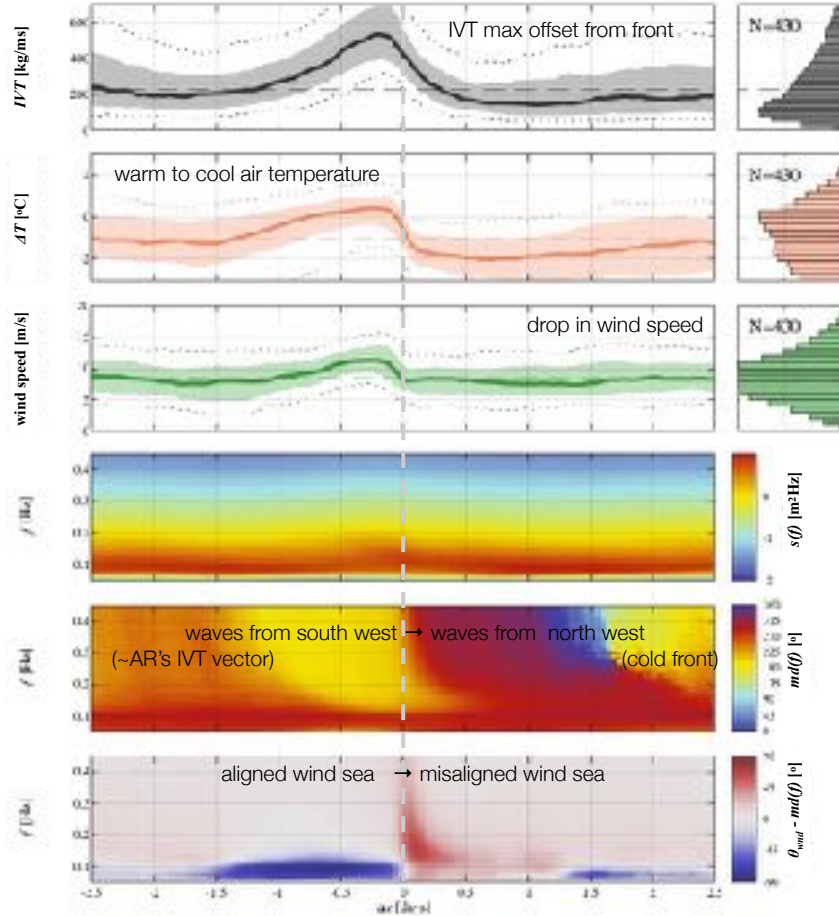
composite wave conditions (4 NDBC buoys) | 2021-2026 (5 years)



composite wave conditions (111 LDL drifters) | 2021-2026 (5 years)



composite wave conditions (106 SOFAR drifters) | 2021-2022 (17 months)



2. evaluating COARE bulk air-sea flux parameterization under atmospheric rivers

Momentum flux in COARE3.5

$$\tau = \rho_a C_D \Delta U^2$$

$$C_D = \left[\frac{\kappa}{\ln\left(\frac{z}{z_0}\right) - \Psi_m\left(\frac{z}{z_0}\right)} \right]^2 + z_0^{\text{rough}}$$

$z_0 = z_0^{\text{smooth}}$

1. Wind Speed Dependent Formulation (WSDF) wind speed only

$$z_0^{\text{rough}} = \alpha \frac{u_*^2}{g} \quad \alpha = f_1(U_{10N})$$

α : Charnock coefficient

2. Wave-Based Formulation (WBF) wave age and wave slope

$$z_0^{\text{rough}} = H_s \cdot 0.09 \cdot \left(\frac{u_*}{C_p} \right)^2$$

significant wave height $\rightarrow$ $\left(\frac{u_*}{C_p} \right)^2$ "peak" wave speed $C_p = g \cdot (T_p / 2\pi)$

Can changing sea states and misalignment under ARs modify surface stress and drag?

Revised wave representation (Sauvage et al. 2023)

3. Revised Wave-Based Formulation (WBF θ) wind-wave alignment

Based on previous studies (e.g. Patton et al. 2015, Porchetta et al. 2019, 2020)

$$z_{\text{rough}} = H_s \cdot 0.09 \cos(0.4\theta) \cdot \left(\frac{u_*}{C_p} \right)^{2 \cos(0.32\theta)}$$

θ : wind direction vs wave direction

4. Revised Wave-Based Formulation (WBF m) mean wave period

Peak wave period does not always represents accurately the sea state, tends to overestimate the impact of mixed sea under moderate wind conditions

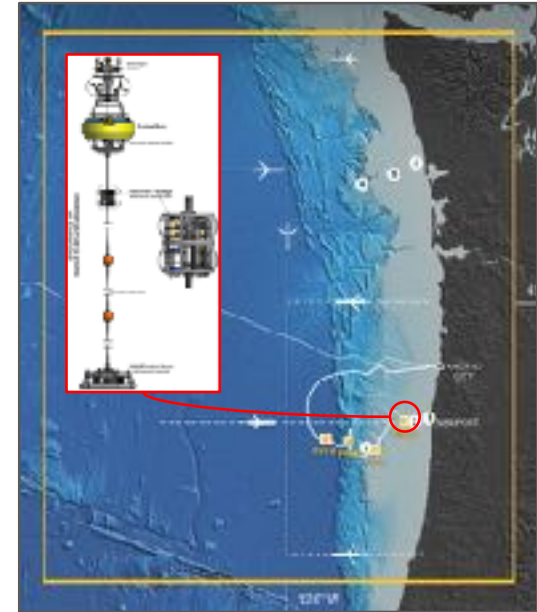
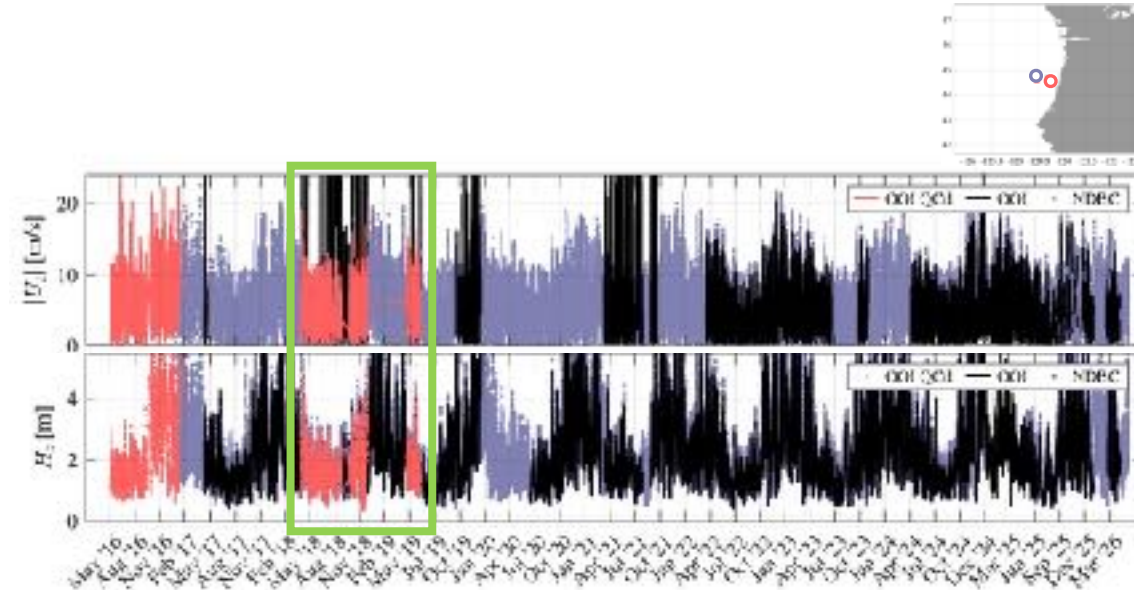
$$z_{\text{rough}} = H_s \cdot 0.39 \cdot \left(\frac{u_*}{C_m} \right)^{2.6}$$

$C_m = g \cdot (T_m / 2\pi)$ "mean" wave speed

2. evaluating COARE bulk air-sea flux parameterization under atmospheric rivers

ONGOING: benchmark against DC measurements

- direct friction velocity measurements requires 3D sonic anemometer integrated with high-frequency IMUs for platform motion correction
- c/o Jim Edson + PMEL for access to QC'd direct covariance data from OOI Endurance Array (Oregon Shelf Surface Mooring) or OSP



2. evaluating COARE bulk air-sea flux parameterization under atmospheric rivers

ONGOING: benchmark against DC measurements

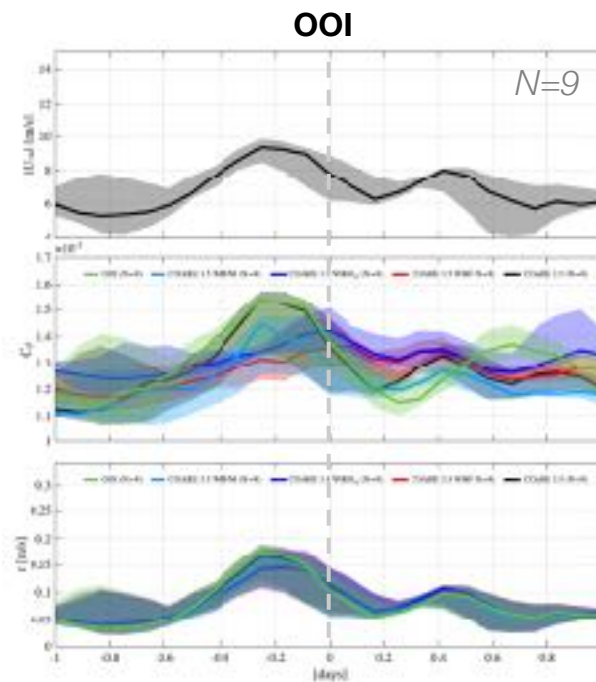
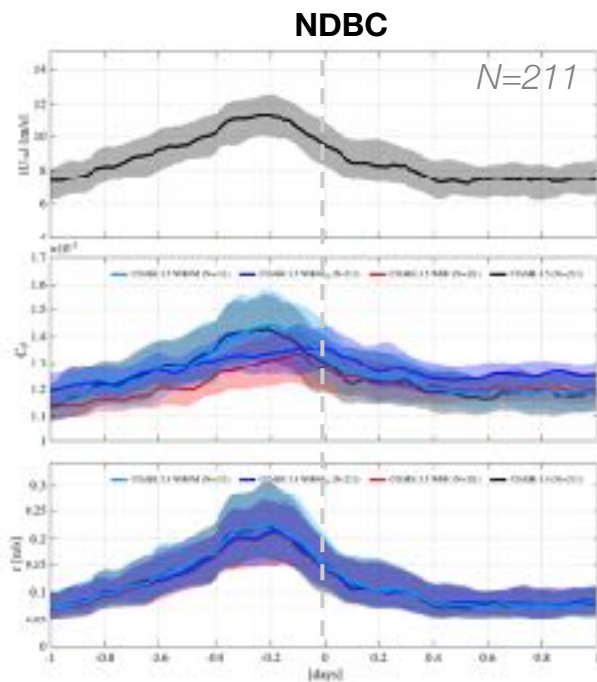
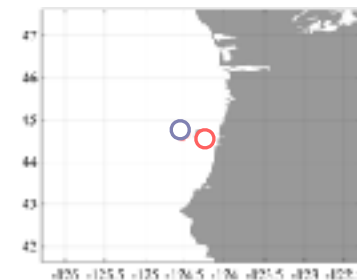
1. Wind Speed Dependent Formulation (WSDF)

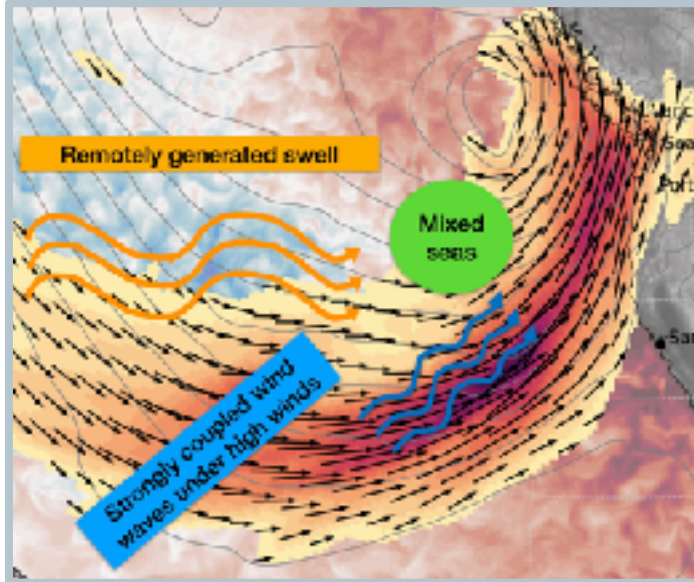
2. Wave-Based Formulation (WBF)

3. Revised Wave-Based Formulation (WBF0)

4. Revised Wave-Based Formulation (WBFM)

Direct-covariance measured





1. Atmospheric cold fronts create persistent wind-wave misalignment within ARs
2. These sea states can lead to differences in surface drag, but representation varies under different bulk flux parameterizations
 - ongoing work to benchmark against DC measurements

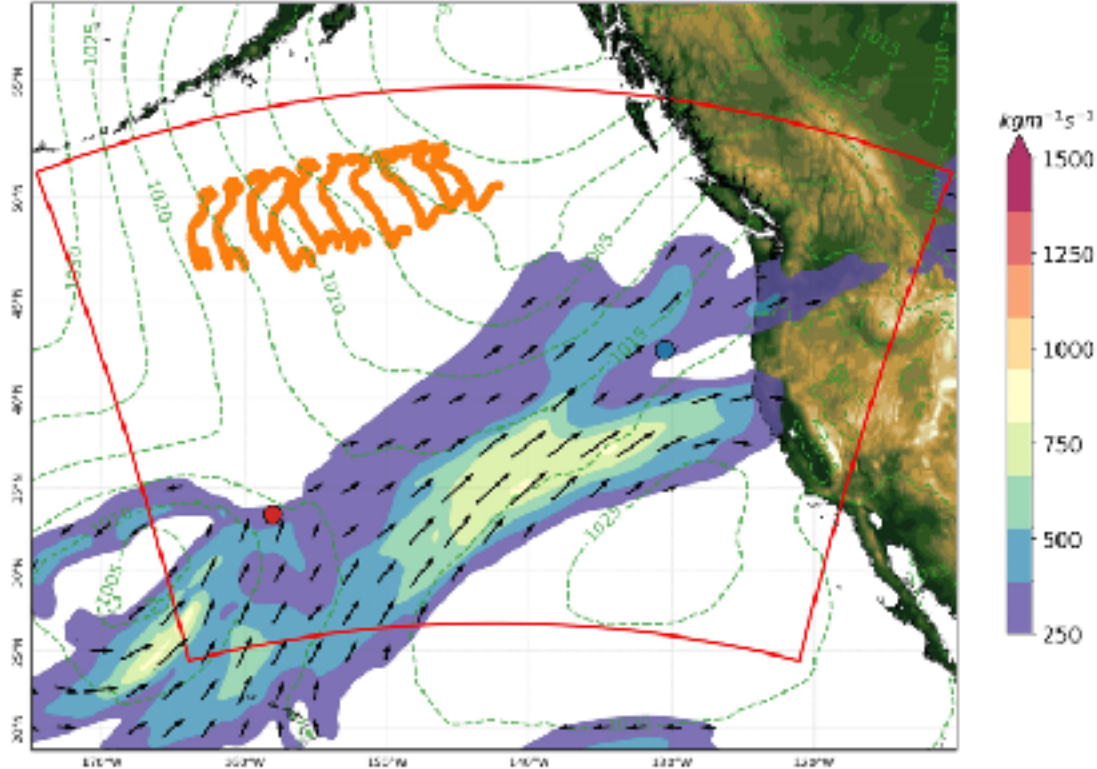
How are these conditions represented in models, and what are the implications for air-sea fluxes?

- validation of the model skill in representing the observed sea states under AR
- source term diagnostics to understand the underlying physics of the sea state evolution

3. model representation of wind-wave interactions in atmospheric rivers

Preliminary Wavewatch 3 simulations

ERA5 IVT on December 16, 10:00 UTC



● SAFARI

● NDBC

● LDL
drifters

WW3 domain

→ 5 km resolution

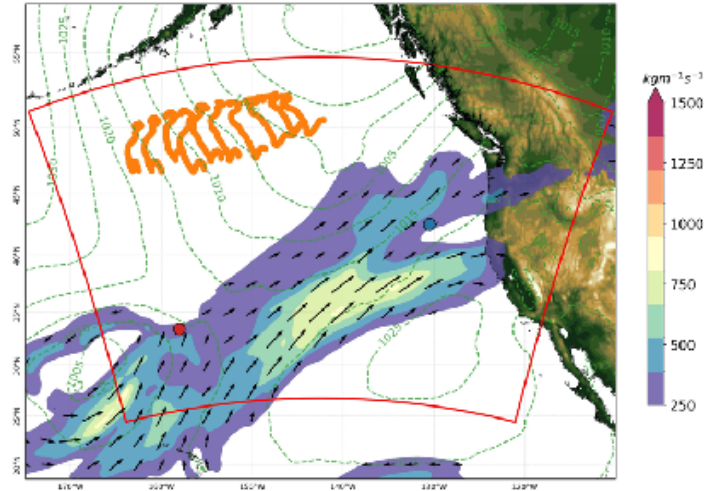
→ 2 months: 11/01-12/31, 2025

→ forced with ERA5 surface winds

→ no ocean surface current

Preliminary Wavewatch 3 simulation

ERA5 IVT on December 16, 10:00 UTC



WW3 domain

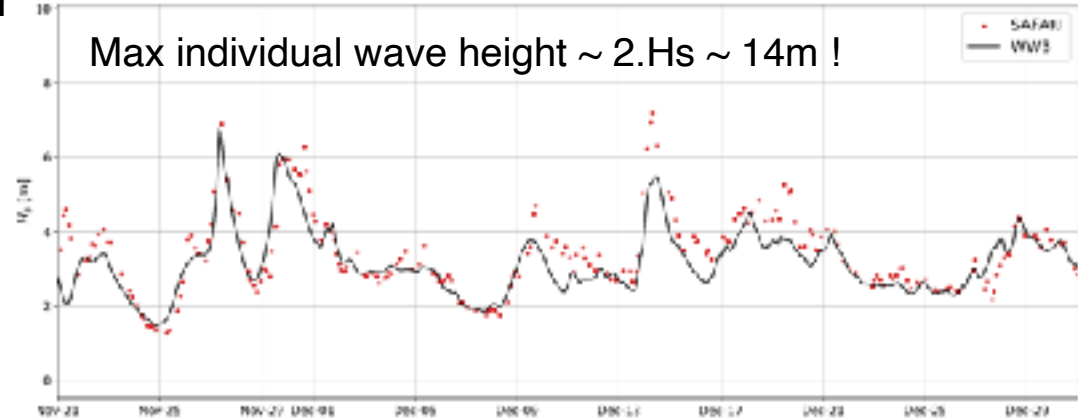
→ 5 km resolution

→ 2 months: 11/01-12/31, 2025

→ forced with ERA5 surface winds

→ no ocean surface current

● SAFARI ● NDBC ● LDL drifters

Max individual wave height $\sim 2.H_s \sim 14\text{m}$!

1 drifter wave height and mean period

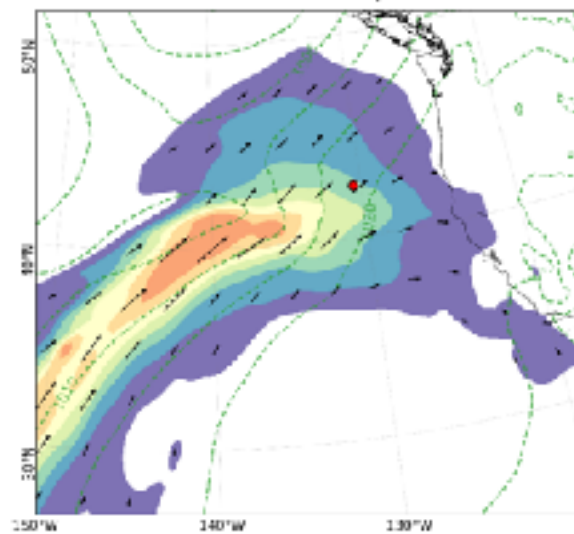


Mean scores:
bias: 0.08m
 r^2 : 0.81
rmse: 0.47m

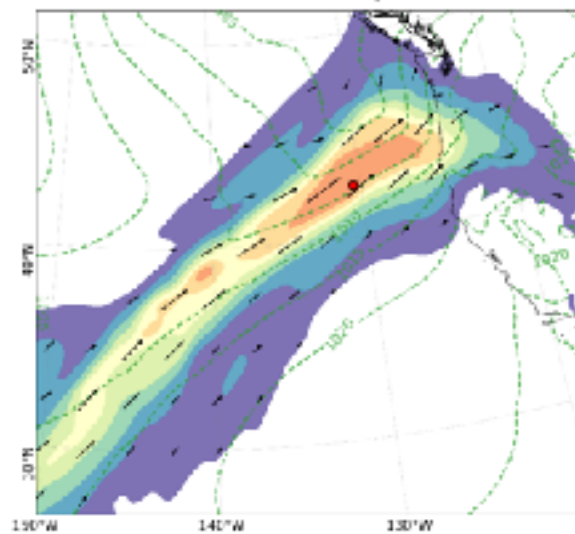


Mean scores:
bias: -0.06s
 r^2 : 0.72
rmse: 0.54s

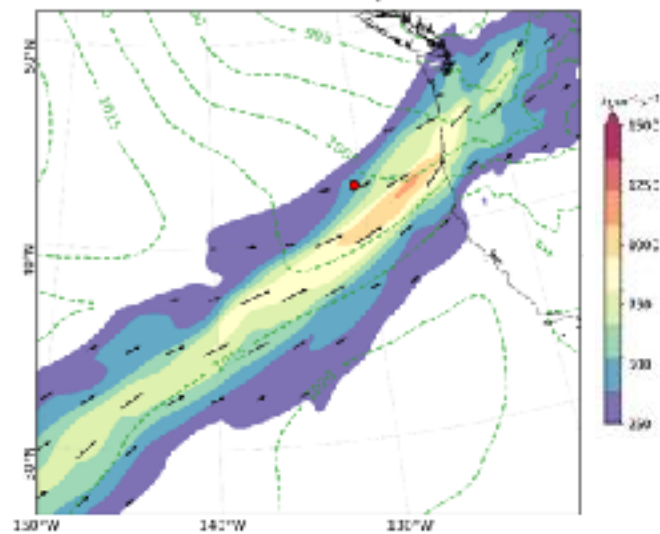
EARS IVT on December 18, 03:00 UTC



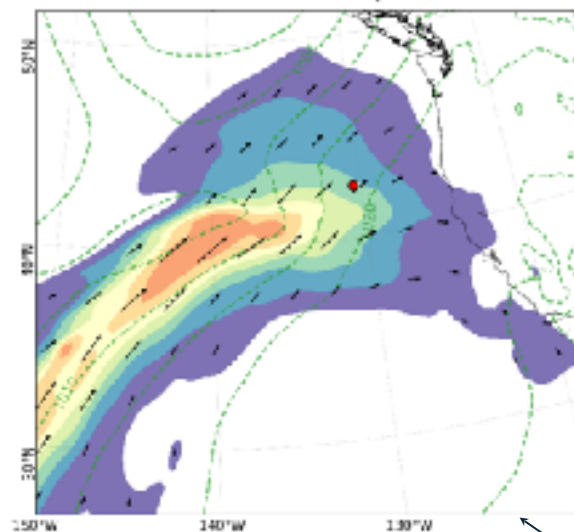
EARS IVT on December 18, 15:00 UTC



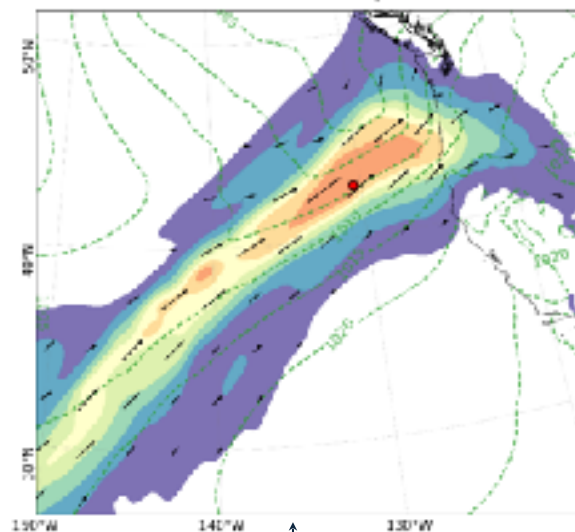
EARS IVT on December 19, 03:00 UTC



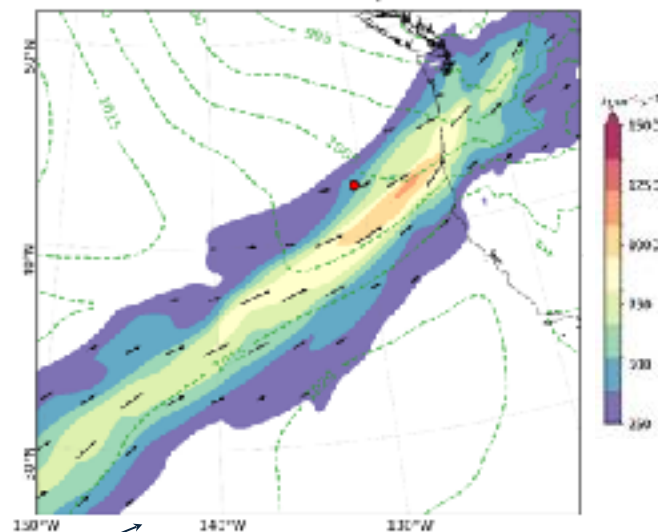
ERA5 IVT on December 18, 03:00 UTC



ERA5 IVT on December 18, 15:00 UTC

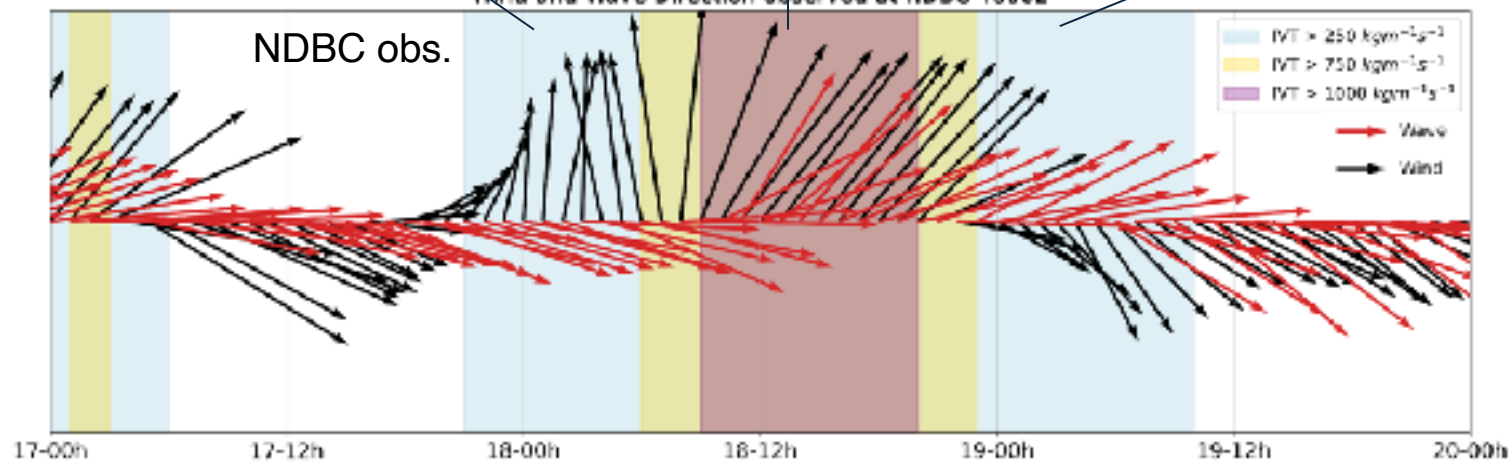


ERA5 IVT on December 19, 03:00 UTC

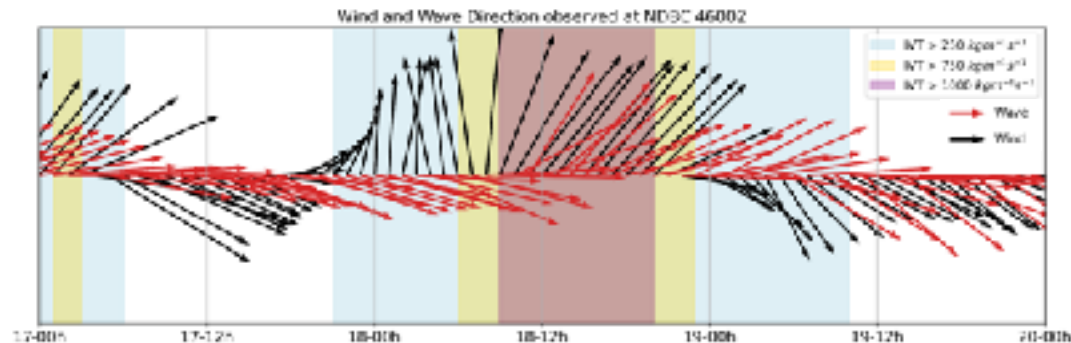


Wind and Wave Direction observed at NDBC 46002

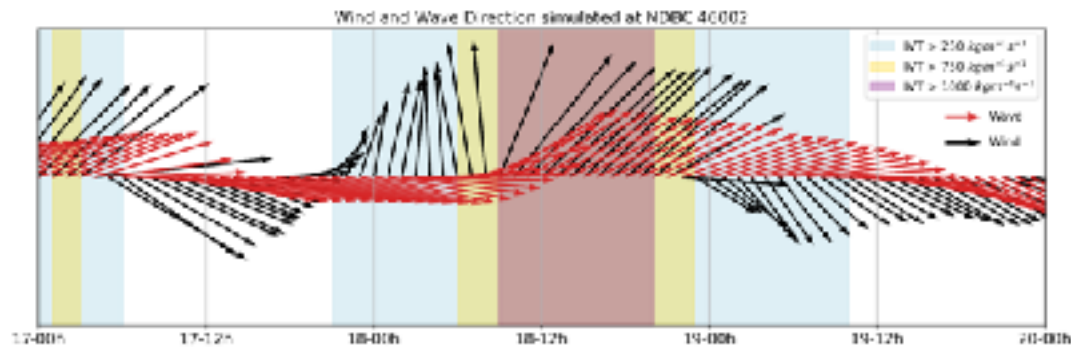
NDBC obs.



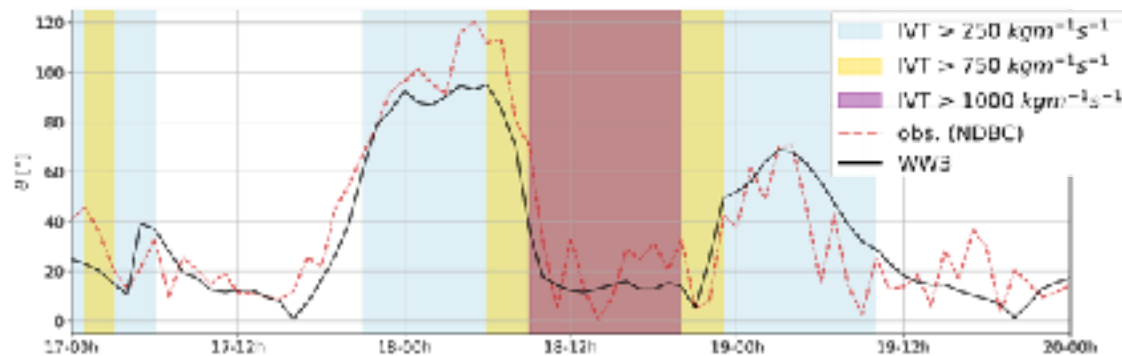
NDBC



WW3 model

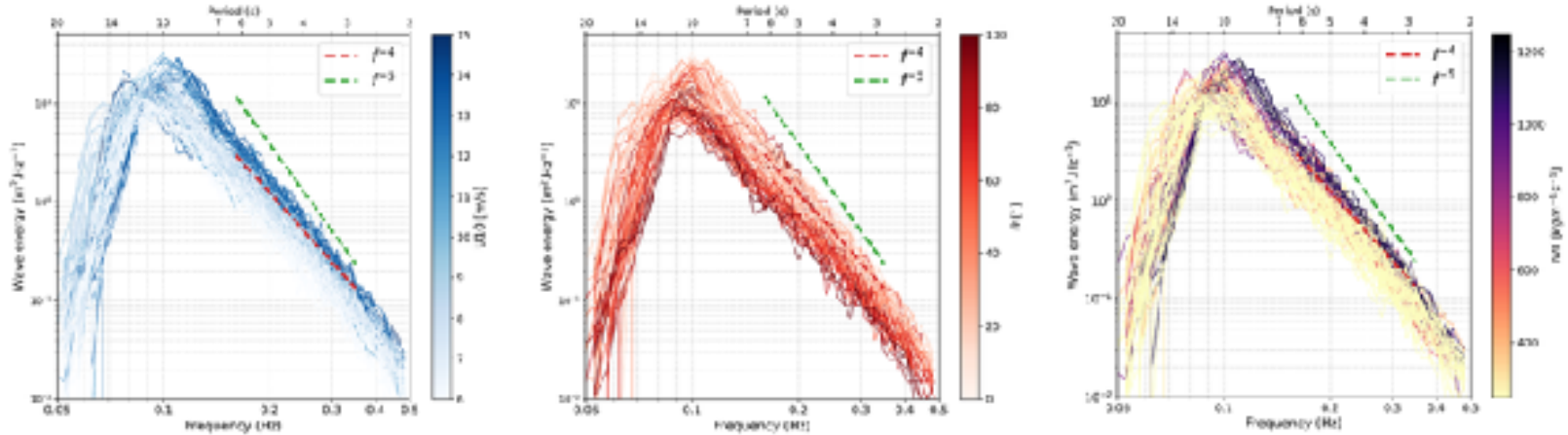


Absolute angle
between wind
and waves

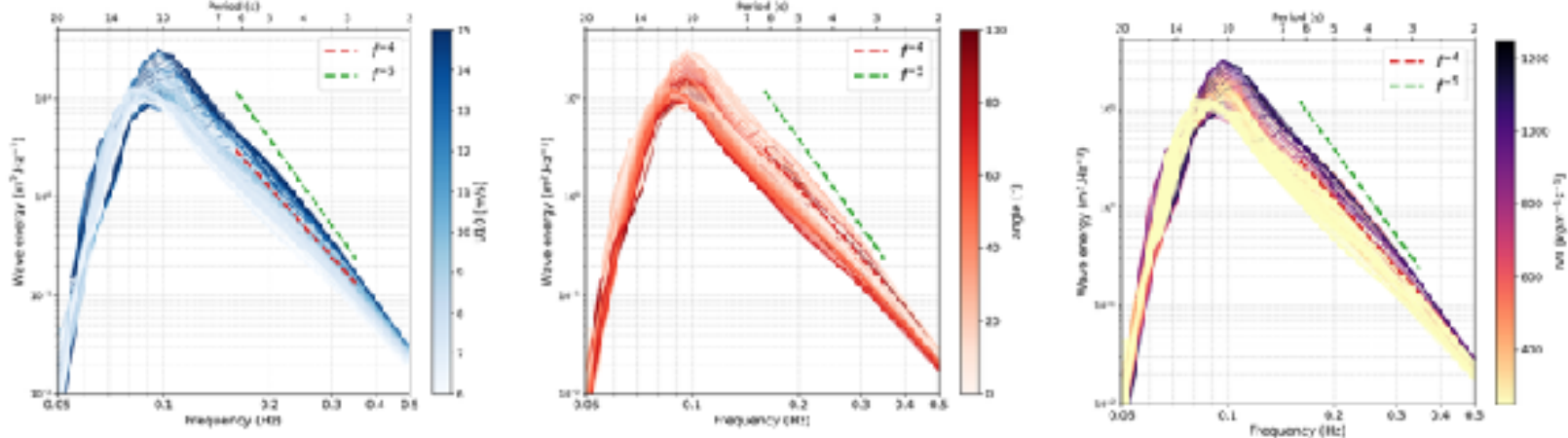


Wave energy spectrum comparison against NDBC

NDBC



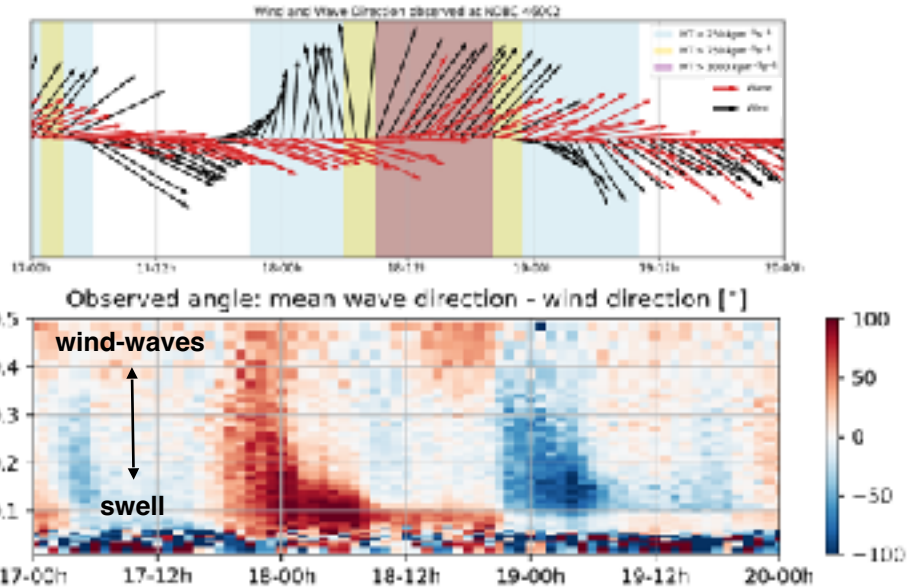
WW3 model



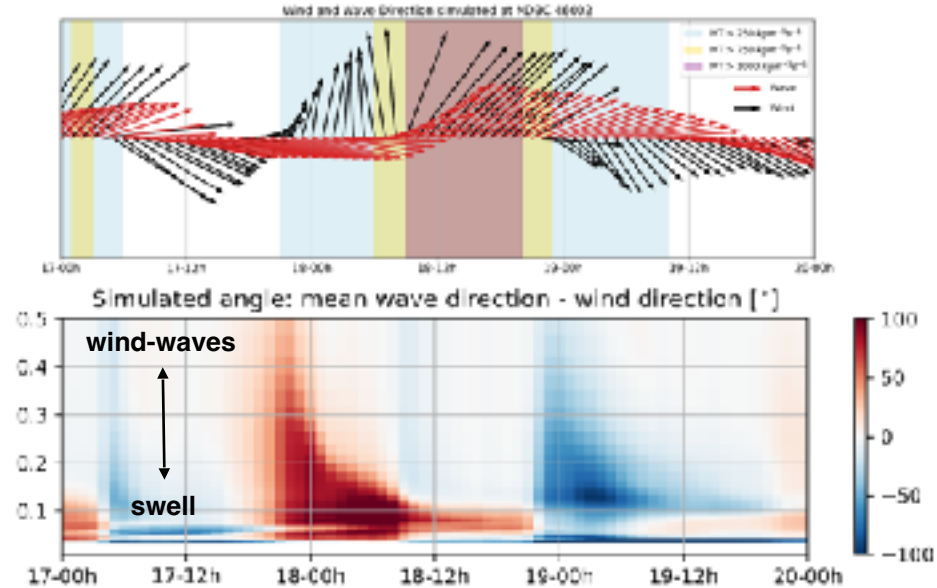
--- f-4 wave equilibrium range --- f-5 wave saturation range

Wave energy spectrum comparison against NDBC

NDBC



WW3 model



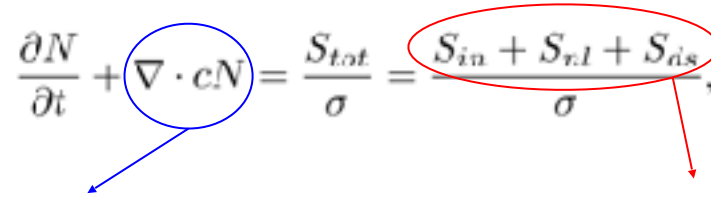
Wave action (N) balance equation

$$\frac{\partial N}{\partial t} + \nabla \cdot cN = \frac{S_{tot}}{\sigma} = \frac{S_{in} + S_{rl} + S_{ds}}{\sigma},$$

Advection term

Physical processes that
modify the wave spectrum

Wave action (N) balance equation

$$\frac{\partial N}{\partial t} + \nabla \cdot cN = \frac{S_{tot}}{\sigma} = \frac{S_{in} + S_{nl} + S_{ds}}{\sigma},$$


Advection term

Physical processes that
modify the wave spectrum

S_{in} = wind input → direct wave growth due to wind

S_{ds} = wave dissipation → swell dissipation and wave breaking

S_{nl} = nonlinear wave-wave interactions → energy transfer within wave spectrum

(Other source terms for wave-ice interactions, depth induced breaking, bottom friction)

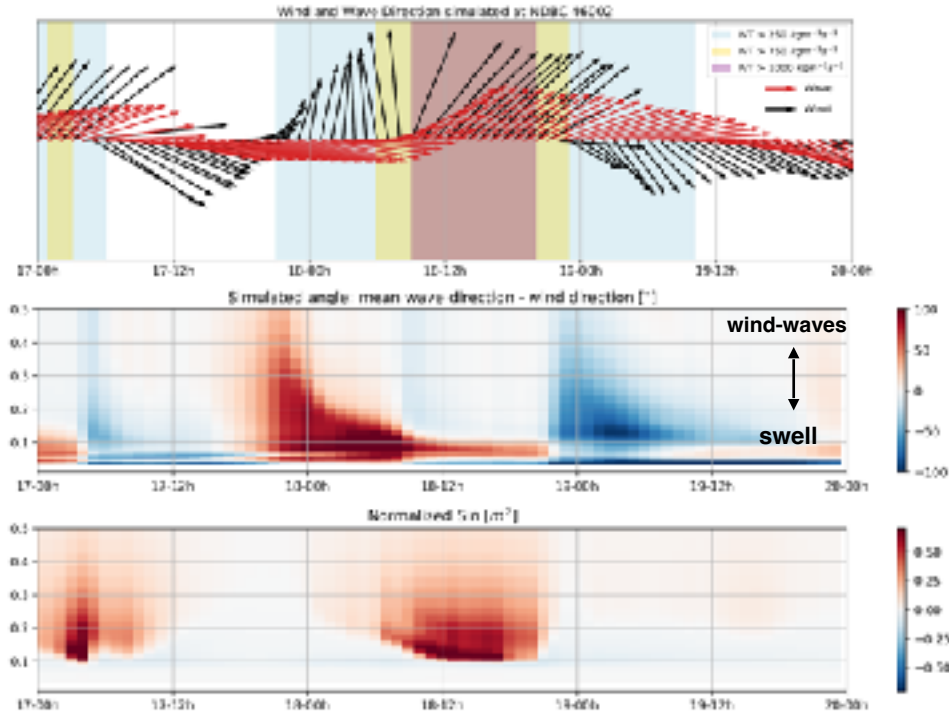
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S_{ds} = wave dissipation

S_{nl} = nonlinear wave-wave interactions



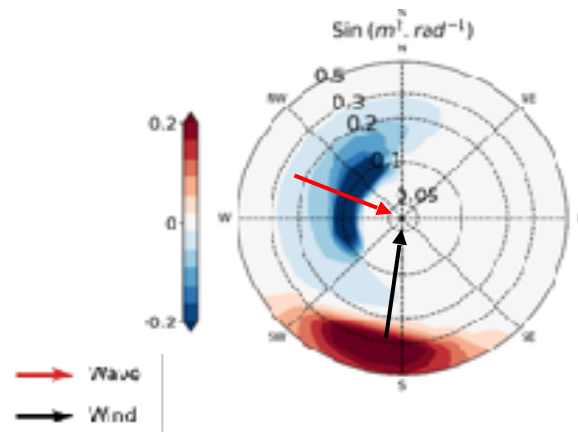
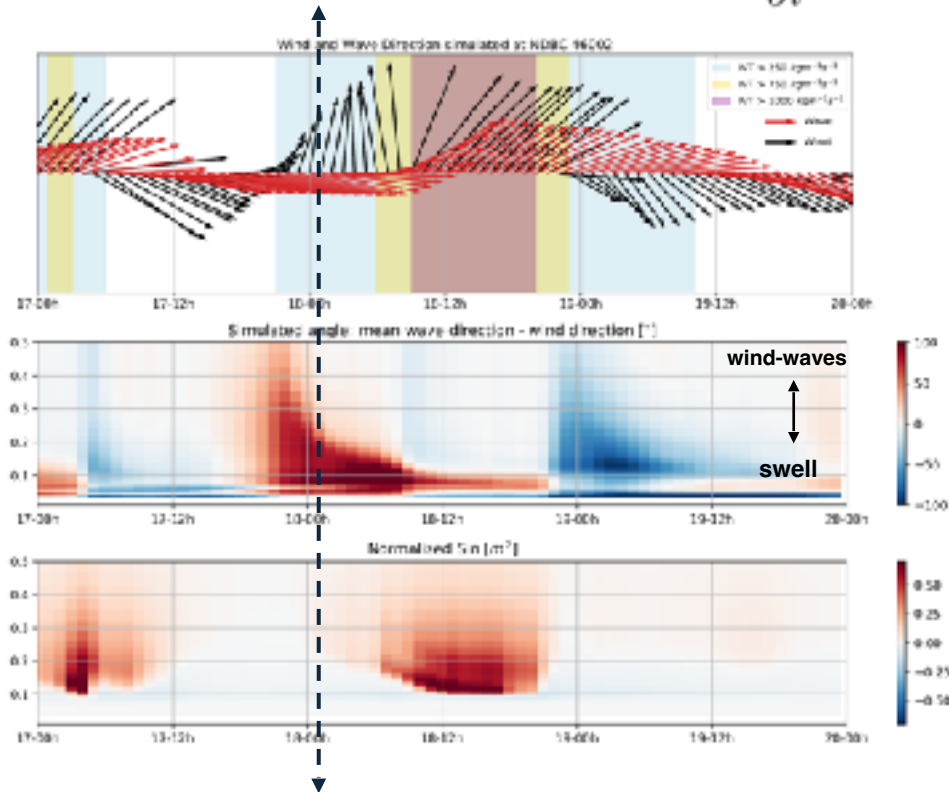
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S_{ds} = wave dissipation

S_{nl} = nonlinear wave-wave interactions



Polar plot center = swell | edge = wind-waves

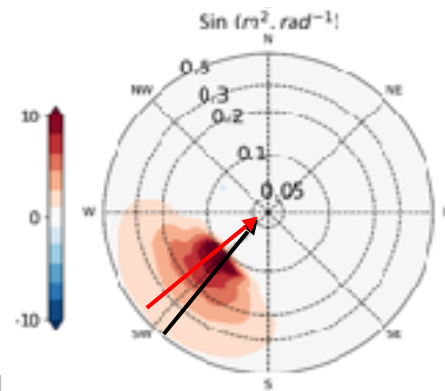
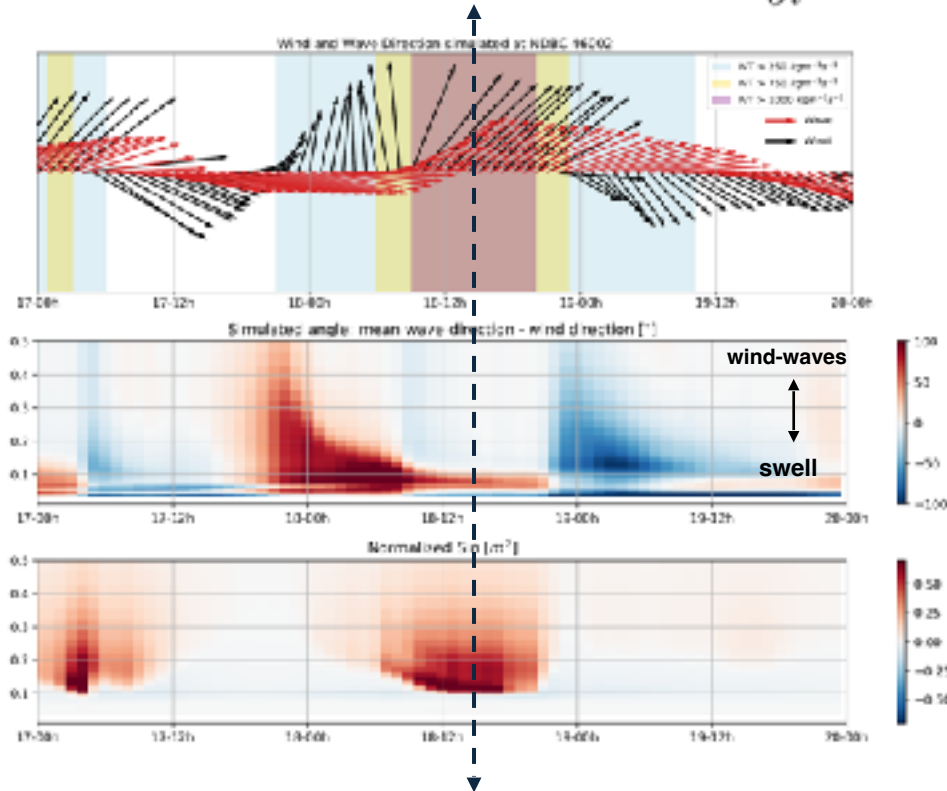
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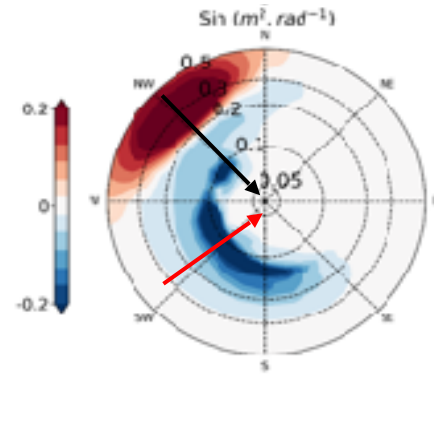
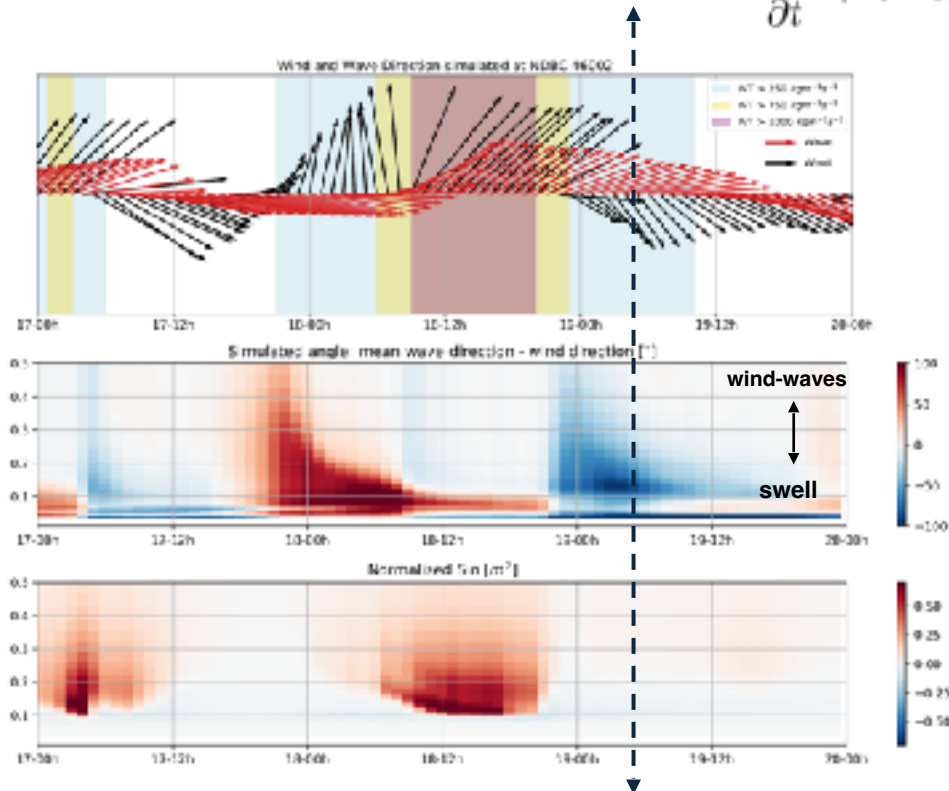
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Polar plot center = swell | edge = wind-waves

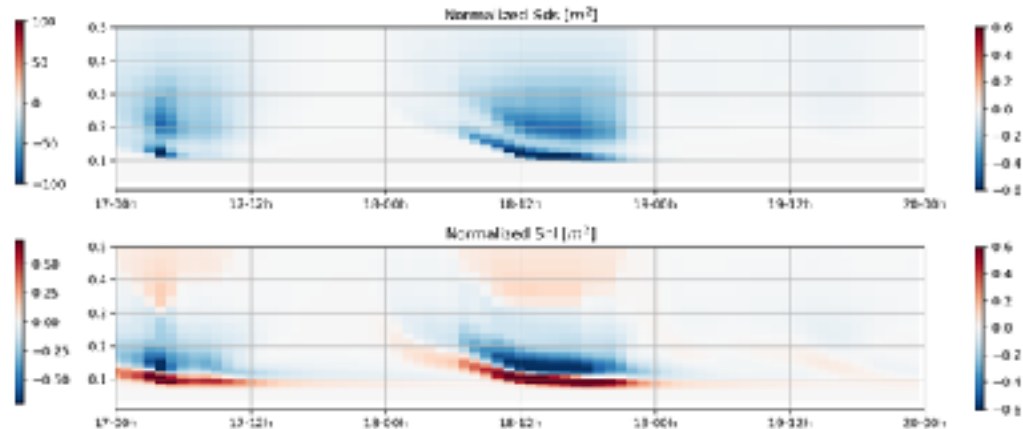
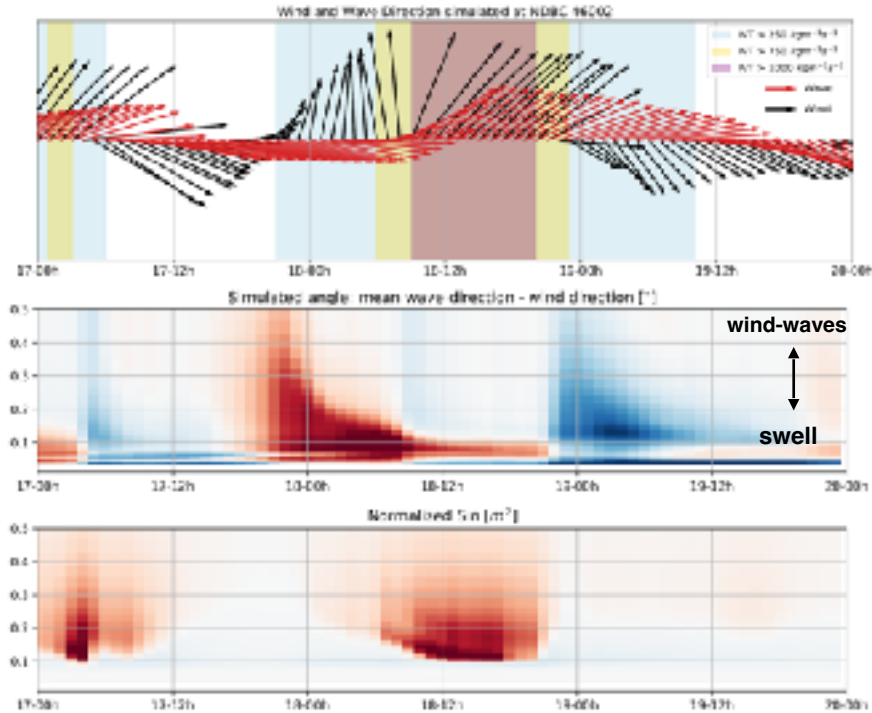
Wave action (N) balance equation

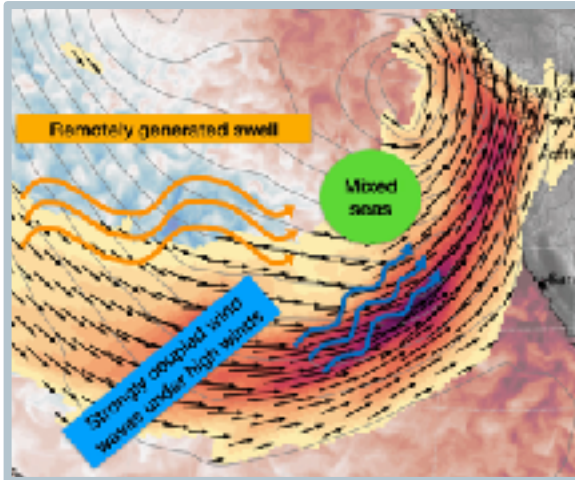
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S_{in} = wind input

S_{ds} = wave dissipation

S_{nl} = nonlinear wave-wave interactions





SUMMARY

1. Atmospheric cold fronts create persistent wind-wave misalignment within ARs
2. These sea states can lead to differences in surface drag, but representation varies under different bulk flux parameterizations
3. Wave model show good performance compared to observations and reproduce the observed sea state under ARs. As the AR approaches, strong wind input progressively align waves from high frequencies to low frequencies

OUTLOOK: Diagnose the impact of wind-wave wave misalignment on surface stress and upper ocean mixing using observations and how to properly represent it in full ocean-atm-waves coupled model